Sahaj Bhadja

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Summary

Device Software Engineer with 2+ years of experience building **C++/Python systems**, **test automation**, and reliable software through OOP, simulation, debugging, documentation, and configuration management.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Tripled throughput across 1K+ AI jobs by restructuring **concurrent backend orchestration** with explicit states, failure isolation, recoverable processing, and observable workflows under sustained production load reliably daily.
- Achieved 99%+ reliability across 5K+ monthly jobs through **production data workflows**, implementing incremental processing, validation, reconciliation, error handling, cloud storage, CI/CD releases, and stable observability reliably daily.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive, repeatable workflows during concurrent model execution and production debugging reliably.
- Maintained 90%+ grading consistency across 10K+ submissions through **structured evaluation workflows**, model switching, fallback paths, persistent results, and instructor review controls supporting reproducible verification reliably at scale.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Improved long-horizon trajectory planning by 18% in simulation by building a **PyTorch decision-transformer pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Raised closed-loop stability 15% through **quantitative verification** of HD maps, ego paths, nearby agents, attention, and distance-aware interactions across diverse autonomous-driving simulations and edge cases repeatedly and reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding failure reports and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows reliably daily.
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments reliably at scale.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellowship *India*

- Reduced centralized-control and quantum-risk exposure 25% in benchmark simulations by designing **consensus algorithms** and evaluating five lattice-based cryptographic primitives across distributed protocol scenarios for peer-reviewed research publication.
- Published peer-reviewed **systems research** and presented at IISc Bengaluru, documenting architecture, simulation methodology, threat models, experimental trade-offs, and reproducible technical findings for distributed consensus design and evaluation.

Skills

Languages: C++, Python, Java, SQL, Bash, Shell, JavaScript
Systems & Design: Object-Oriented Design, Data Structures, Algorithms, Operating Systems, Concurrency, Multithreading, Process Scheduling, Distributed Systems
Testing & Verification: Unit Testing, Integration Testing, Automated Testing, PyTest, Mockito, Simulation, Debugging, Data Validation, Error Handling
Tooling & Configuration: Git, GitHub, GitHub Actions, Jenkins, CI/CD, Docker, Linux, Windows
Engineering Practices: Technical Documentation, System Design, Failure Isolation, Root-Cause Analysis, Performance Optimization, Requirements Analysis, API Design, Cross-Platform Development

Projects

Embedded Device Communication Test Harness - Design Concept

- Designed a **C++/Python device-test harness concept** supporting configurable serial-interface adapters, command framing, timing controls, simulated devices, fault injection, structured logs, deterministic test cases, and reproducible integration diagnostics.

Regulated Software Traceability Toolkit - Design Concept

- Designed a **software traceability toolkit concept** linking requirements, design descriptions, source revisions, risk controls, test cases, execution evidence, defects, and technical reports through versioned records and coverage checks.

CPU Scheduling Simulator | Live

- Implemented **C++ scheduling algorithms** for FCFS, SJF, SRT, Round Robin, priority, and MLQ, with edge-case handling, process-state visualization, and dynamic turnaround and waiting-time calculations across simulations reliably.